OTT STREAMING CONTENT TREND ANALYSIS

Amazon Prime Video Catalogue | Project 2 of 5

Data Analytics Internship – Veda Technology

1. Project Overview

This project examines the supplied Amazon Prime Video catalogue dataset to understand its content mix and metadata patterns. Python and Pandas are used for data preparation and exploratory analysis, SQL provides reusable queries for catalogue analysis, and Excel presents cleaned records and summary tables with charts.

2. Objective

The objective is to explore the distribution of movies and TV shows, identify frequently listed genres and countries, review content ratings and release-year patterns, examine movie runtime and TV season counts, and identify data-quality limitations that should be considered before making business decisions.

3. Dataset Description

The supplied file contains 9,668 records and 12 columns. Fields include show ID, content type, title, director, cast, country, date added, release year, rating, duration, listed genres and description. The latest release year represented is 2021. This is a catalogue metadata snapshot, not a live view of the current Amazon Prime catalogue.

4. Dataset Profile

Measure	Value
Total titles	9,668
Movies	7,814
TV shows	1,854
Duplicate show IDs	0
Rows with at least one missing field	9,667
Most common release year	2021 (1,442 titles)

5. Data Preparation

The analysis checks duplicate show identifiers and missing values. Text fields used for analysis are handled consistently, date-added values are parsed as dates, and numeric duration values are extracted from strings such as movie minutes or TV seasons. Genre and country fields can contain multiple comma-separated values; these are split into separate entries for multi-label summaries. A title may therefore contribute once to each genre or country listed for it.

6. Key Catalogue Findings

Content type: The dataset contains 7,814 movies and 1,854 TV shows. Movies make up 80.8% of the records, while TV shows make up 19.2%.

Genre: Drama is the most frequently listed genre, appearing against 3,687 title-genre combinations. Genre counts are multi-label counts, so a title listed in several genres is counted in each applicable genre.

Country: United States is the leading country entry with 334 titles in the exploded country summary. Some titles list multiple countries, and country metadata is missing for a substantial portion of the catalogue.

Content rating: 13+ is the most common rating label, with 2,117 records. Rating labels describe catalogue classification and should not be interpreted as audience ratings or review scores.

Release year: The most represented release year is 2021, with 1,442 titles. The maximum release year in this supplied snapshot is 2021.

Top Five Listed Genres

Genre	Titles
Drama	3,687
Comedy	2,099
Action	1,657
Suspense	1,501
Kids	1,085

7. Visual Analysis Included

The Jupyter Notebook includes visualizations for movie versus TV-show counts, top genres, content ratings, release-year distribution, top countries, movie runtime distribution, and titles added to the catalogue by year. The Excel workbook contains cleaned title data and category summary sheets with charts for key catalogue dimensions.

8. Content Strategy Recommendations

1. Review Drama as a core catalogue category. Drama is the most frequently listed genre in this dataset. The content team can review whether new drama acquisitions fill specific language, subgenre or regional gaps. Title counts should be assessed alongside viewing hours, completion rates, subscriber acquisition and licensing costs.

2. Evaluate regional catalogue depth. The country summary can help identify markets represented in the catalogue and potential gaps for local-language content or co-productions. Because country fields are frequently missing and may list several countries, the findings should be treated as directional until metadata is improved.

3. Review the balance of movies and episodic content. Movies substantially outnumber TV shows in this snapshot. The platform could investigate whether selected genres and markets need more series, using repeat viewing, episode completion and retention data before making acquisition decisions.

9. SQL Analysis

The accompanying MySQL 8+ script includes queries for total titles, content type, release year, ratings, genre groupings, country values, date-added trends, movie runtime, TV seasons, type-by-rating summaries, missing metadata, primary genres and directors. The CSV must first be imported into a MySQL table named amazon_prime_titles.

10. Limitations

This dataset does not include actual streams, viewing hours, audience engagement, subscriber behavior, revenue, licensing costs or profitability. Therefore, catalogue counts cannot prove audience popularity or commercial success. Multi-value genre and country fields require careful interpretation. Missing metadata may affect comparisons. Ambiguous date formats should be checked against the source convention before analysis.

11. Conclusion

The project transforms catalogue metadata into structured summaries that can support content planning discussions. It demonstrates a practical workflow across Python, SQL and Excel, while identifying the limits of metadata-only analysis. Combining this catalogue with viewing and commercial data would enable stronger acquisition and retention recommendations.

Prepared for Veda Technology | Project 2 of 5 – Data Analytics Internship